CPE 381 Classwork 12: Fourier Transform

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: October 14, 2026 (In-Class)

1. Fourier transform of a signal $x(t)$ can be calculated as

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt$$

Based on the above formula calculate the Fourier transform of the following: **(10 Points)**

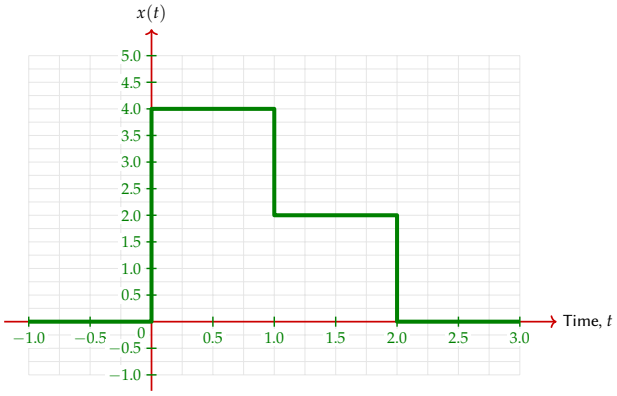


Figure 1: Staircase signal $x(t) = 4\,[u(t) - u(t-1)] + 2\,[u(t-1) - u(t-2)]$.

Answer

2. Find the fourier transform of the triangle function show below: **(10 Points)**

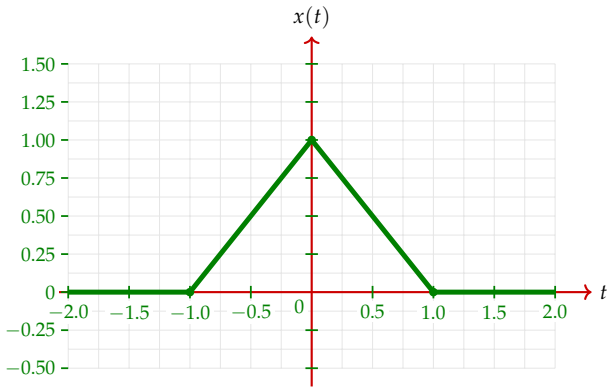


Figure 2: Triangle Fnction

Answer